ABSTRACT

The Mobile Manufacturing System is an Oracle-based relational database mini project for maintaining and analyzing information related to mobile brands, models, components, suppliers, manufacturing, inventory, sales, markets, e-commerce platforms, promotions and third-party promoters. The system converts the broad institutional requirement into a manageable database-focused solution.

The database records the components required by each mobile model and their unit costs, production batches and finished inventory, and sales through India and international markets and different e-commerce platforms. Revenue and profitability are derived through SQL rather than unnecessarily duplicating calculated values.

Promotional campaigns and third-party promoters such as YouTubers/influencers are represented using dedicated tables. The selected novelty is Component-Cost Impact Analysis, which identifies the contribution of individual components to a model's total component cost and links manufacturing cost to profitability.

The project demonstrates relational design, normalization to approximately 3NF, constraints, joins, aggregation, subqueries, views, sequences, procedures, functions and triggers using Oracle SQL/PLSQL.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Mobile manufacturing involves interconnected activities such as component procurement, supplier selection, production, stock management, sales, and promotion. When these activities are maintained through separate records, it becomes difficult to obtain a consolidated view of component costs, production information, inventory, sales performance, and profitability.

1.2 Interpretation of the Original Abstract

The original Mobile Manufacturing System abstract requires a database that can provide information about mobile brands and models, the parts required and their cost, manufacturing activities, sales in India and abroad through different e-commerce platforms, profit based on sales, and promotional activities involving third-party promoters such as YouTubers. If interpreted as a complete business application, this would require supply-chain management, procurement, logistics, accounting, real-time e-commerce integration, and social-media integration. Such a system would be too large for a DBMS mini project. Therefore, the proposed project refines the abstract into a focused relational database and business-analysis system.

1.3 Need for the Proposed System

- Centralized storage of mobile brands, models, components, suppliers, production, inventory, and sales data.
- Traceability from component requirements and sourcing to manufactured mobile models and sales.
- Identification of the components required for each mobile model and their associated costs.
- Manufacturing cost, revenue, and profit analysis through SQL queries.
- Comparison of sales performance across Indian and international markets.
- Analysis of sales across different e-commerce platforms.
- A practical method to identify components with high contribution to a model's manufacturing cost.

1.4 Scope of the Project

The project is organized into four integrated modules: (1) Mobile & Component Management, (2) Supplier & Sourcing Management, (3) Manufacturing & Inventory Management, and (4) Sales & Business Analysis. Together, these modules cover mobile product information, component requirements, supplier information, production batches, inventory, sales, profit analysis, e-commerce performance, and promotional information.

The project uses fictional/sample data for demonstration and does not connect directly to Amazon, Flipkart, YouTube, or other external services. The project excludes real-time e-commerce APIs, online payments, customer management, logistics tracking, demand forecasting, machine learning, automated social-media analytics, and complete ERP functionality.

CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Statement

Mobile manufacturers require a structured mechanism for maintaining information about mobile brands, models, components, suppliers, manufacturing activities, inventory, and sales. The cost of components can significantly influence the overall manufacturing cost of a mobile model. When product, component, production, sales, and promotional information is not integrated, it becomes difficult to determine which models are cost-intensive, which models perform well in different markets, which e-commerce platforms generate higher sales, and how revenue compares with manufacturing cost.

The proposed system addresses this problem by providing a centralized Oracle database that connects mobile models with their components, suppliers, production batches, inventory, sales channels, markets, e-commerce platforms, promotions, and third-party promoters. The system supports SQL-based cost, revenue, profit, sales, and promotional analysis.

2.2 Objectives

- Maintain information about mobile brands and their models.
- Maintain a master database of mobile components/parts.
- Maintain supplier information and component sourcing details.
- Record the components and quantities required to manufacture each mobile model.
- Record component unit costs and calculate model-wise component cost.
- Track manufacturing batches and quantities produced.
- Maintain current finished-mobile inventory.
- Record sales through multiple e-commerce and sales channels.
- Distinguish sales in India from international sales.
- Calculate model-wise and brand-wise revenue and profit.
- Analyze sales performance across different e-commerce platforms.
- Maintain promotional campaigns and third-party promoter information.

2.3 Existing System

A simplified existing approach may rely on spreadsheets, manual registers, and disconnected files. Brand and model information, component records, supplier details, production records, stock, sales information, and promotional records may be maintained separately. This can lead to data redundancy, inconsistent updates, difficulty in tracing relationships, and manual calculation of cost, revenue, and profitability.

2.4 Proposed System

The proposed system integrates the mobile manufacturing lifecycle through four consolidated database modules rather than treating every business activity as a separate module. Relational tables and foreign keys connect brands, mobile models, components, suppliers, production batches, inventory, markets, e-commerce platforms, sales, promotions, and promoters. SQL queries and views provide analytical reports, while selected PL/SQL procedures, functions, and triggers implement useful database business logic.

2.5 Novelty / Unique Feature

Component Cost Impact Analysis is the project's principal novelty. A mobile model can require many different components, and each component may have a different unit cost and required quantity. The system calculates the cost contribution of each component by considering its required quantity and unit cost, ranks components according to their contribution, and compares this information with model-wise sales revenue and derived profitability. This makes the project a database-based decision-support system rather than only an inventory or manufacturing management system.

CHAPTER 3: FUNCTIONAL REQUIREMENTS – MODULES AND SOFTWARE USED

3.1 Modules of the System

MODULE	OBJECTIVE	TABLES	OUTPUT
1. Mobile & Component Management	Maintain mobile brand information, model details, component details, and the component quantities required to manufacture each model.	BRAND, MOBILE_MODEL, COMPONENT, MODEL_COMPONENT	Mobile master list, component master list, and model-wise component requirement and cost report.
2. Supplier & Sourcing Management	Maintain suppliers, component sourcing information, country of origin, source type, availability, and cost.	SUPPLIER, SUPPLIER_COMPONENT	Supplier-wise component report, local versus imported sourcing report, and component cost comparison.
3. Manufacturing & Inventory Management	Track mobile production batches, production quantities, manufacturing costs, and current finished-mobile stock.	MANUFACTURING_BATCH, INVENTORY	Manufacturing history, batch/production cost report, current inventory, and low-stock report.
4. Sales & Business Analysis	Record sales through different channels and generate revenue, profit, market, e-commerce, and promotional analysis using related data.	MARKET, ECOMMERCE_PLATFORM, SALE, PROMOTION, PROMOTER	Channel-wise sales, model-wise revenue, profit analysis, most profitable models, India versus international sales, and component-cost impact reports.

3.2 Functional Requirements

- Module 1: Add, update, and retrieve mobile brand, model, and component records.
- Module 1: Define and maintain model-wise component requirements.
- Map multiple components to a mobile model through a model-component relationship.
- Module 2: Record suppliers, component sourcing information, source type, country of origin, availability, and cost.
- Module 3: Record manufacturing batches, production quantities, manufacturing costs, and current inventory.
- Module 4: Record sales through multiple channels and generate revenue and profit reports.
- Module 4: Distinguish India and international sales and generate e-commerce platform analysis.
- Module 4: Maintain promotional campaigns and third-party promoter information.
- Module 4: Generate Component Cost Impact Analysis using related product, component, production, and sales data.

3.3 Non-Functional Requirements

- Data integrity through primary keys, foreign keys, and CHECK/UNIQUE constraints.
- Normalized design to reduce redundancy.
- Implementation using Oracle SQL and PL/SQL.
- Simple and maintainable structure suitable for academic demonstration.
- No dependence on external APIs or advanced infrastructure.

3.4 Hardware and Software Requirements

CATEGORY	REQUIREMENT
Operating System	Windows or another OS supported by Oracle development tools
Database	Oracle Database / Oracle XE where appropriate
Development Tool	Oracle SQL Developer or equivalent
Languages	SQL and PL/SQL
Documentation	Microsoft Word or compatible software
Hardware	Standard laptop/desktop capable of running Oracle development tools

3.5 Overall System Workflow

Brand / Company → Mobile Model → Components & Suppliers → Model Component Requirement → Manufacturing Batch → Inventory → Sales Channel & Sale → India / International Market Analysis → Revenue & Profit Analysis →

Promotions & Third-party Promoters → Component Cost Impact Analysis.

The novelty feature is integrated primarily into Module 1 (Mobile & Component Management) and Module 4 (Sales & Business Analysis): component requirement and cost data identify high-cost components, while sales and revenue data help evaluate their relationship with model profitability.